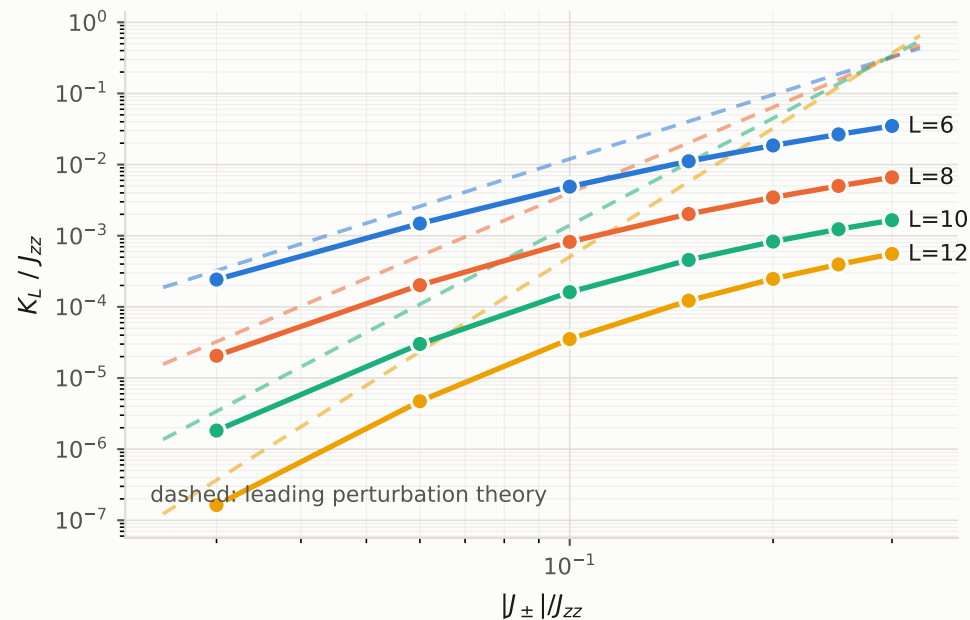


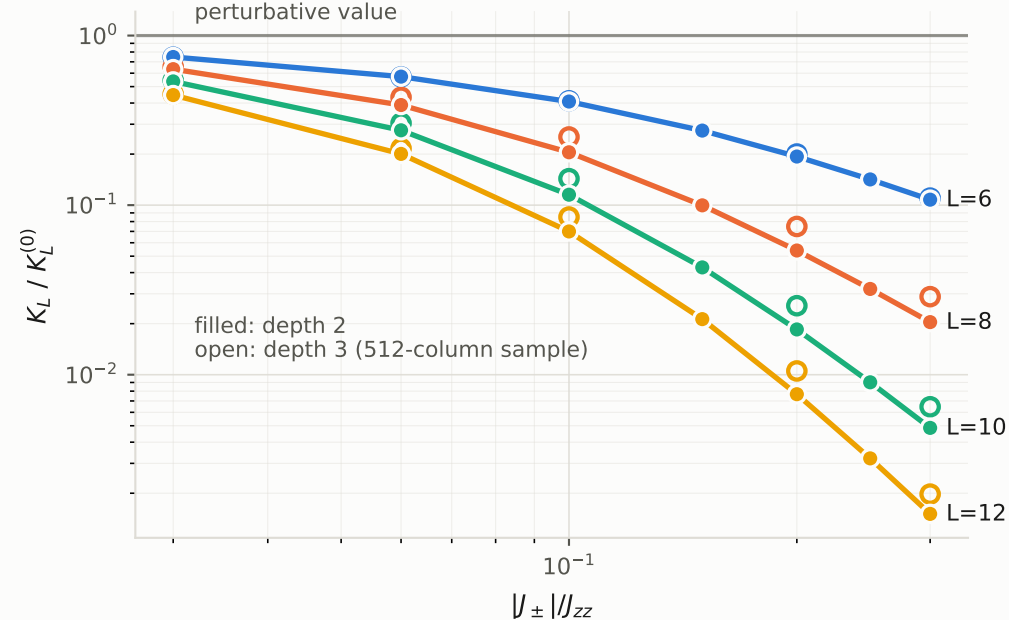
Nonperturbative loop amplitudes of quantum spin ice

—●— $L = 6$ —●— $L = 8$ —●— $L = 10$ —●— $L = 12$

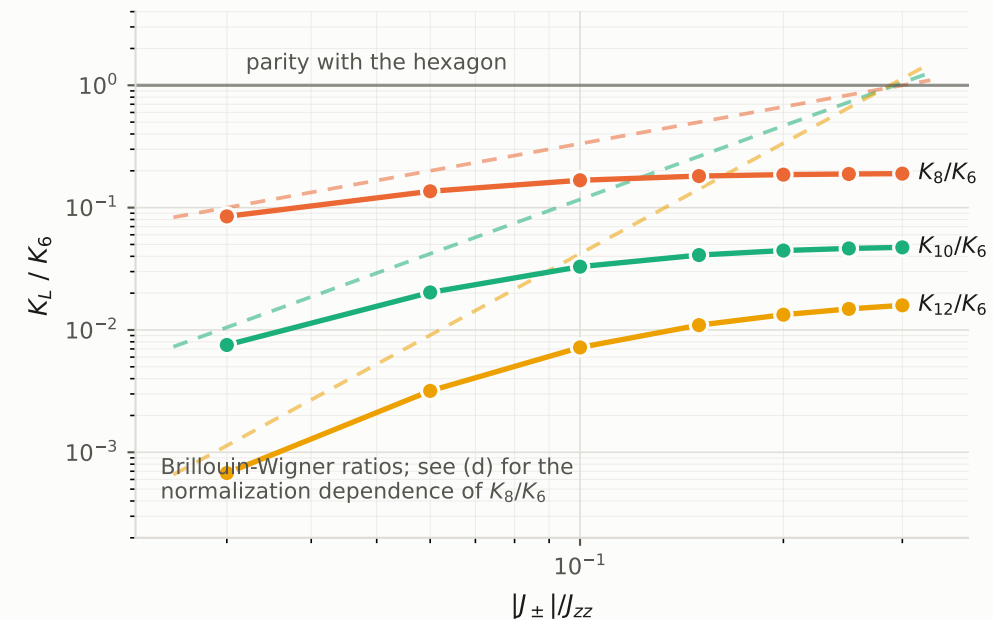
(a) FCC-32 loop amplitudes, virtual depth 2



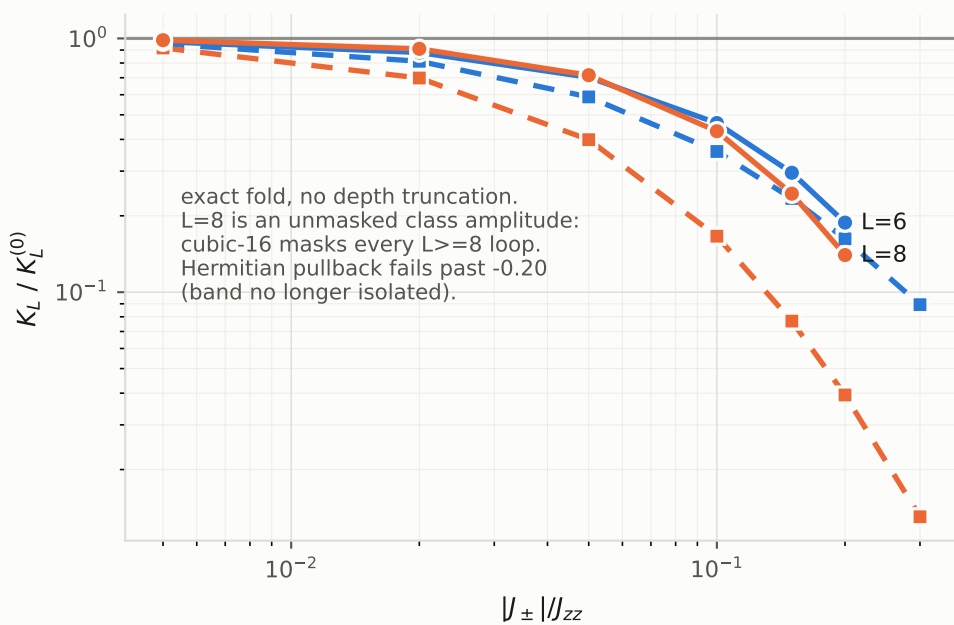
(b) Suppression below perturbation theory



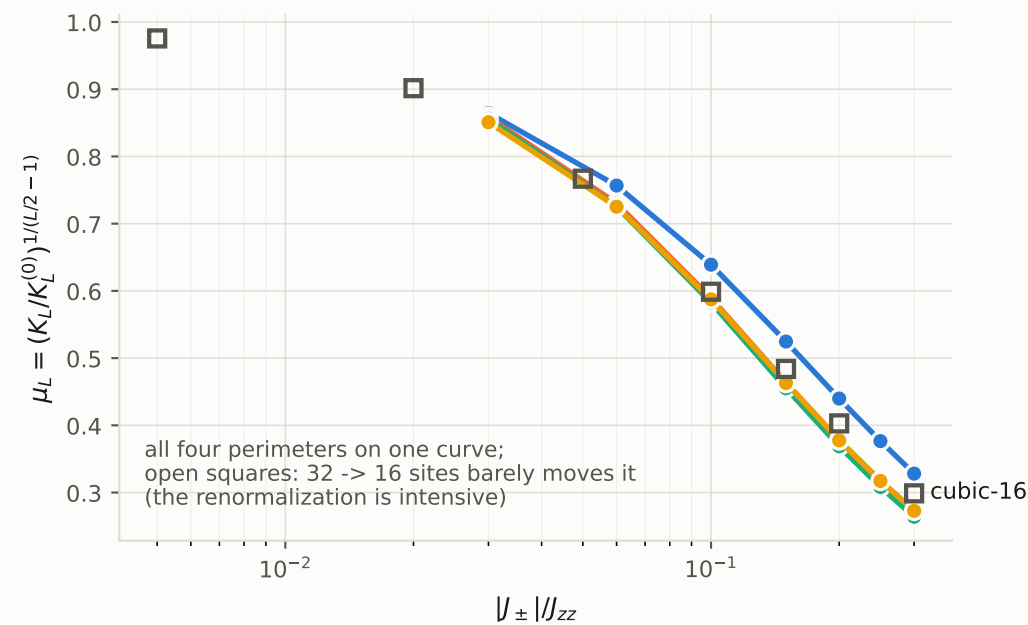
(c) Hierarchy relative to the hexagon



(d) cubic-16: Hermitian (solid) vs Brillouin-Wigner (dashed)



(e) One parameter collapses every perimeter



(f) Residual: the hexagon is protected

